

Cosmological Perturbations in Synchronous and Newtonian Gauges: Formalism and Applications

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Contents

1	Introduction	4
2	The FLRW Background and Cosmological Symmetries	6
2.1	FLRW Metric in Conformal Time	6
2.2	Perfect Fluid Energy-Momentum Tensor	6
2.3	Background Einstein Equations	6
2.4	Equation of State and Cosmological Epochs	7
2.5	Relevance to Perturbation Theory	9
3	Classification of Perturbations and Metric Decomposition	9
3.1	General Metric Perturbation Structure	9
3.2	Metric Perturbation Decomposition: Scalar, Vector, and Tensor	9
3.3	Harmonic Expansion in Flat Space	10
3.4	Fourier Decomposition of Perturbations	11
3.5	Interpretation of the SVT Modes	11
4	The Synchronous Gauge	11
4.1	Gauge Conditions and Metric Form	11
4.2	Residual Gauge Freedom	12
4.3	Perturbed Einstein Equations in Synchronous Gauge	12
4.4	Advantages and Limitations	13
5	The Conformal Newtonian Gauge	13
5.1	Gauge Conditions and Metric Form	13
5.2	Physical Interpretation	14
5.3	Perturbed Einstein Equations in This Gauge	14
5.4	Advantages of the Gauge	14
6	Gauge Transformations and Gauge-Invariant Variables	15
6.1	First-Order Gauge Transformations	15
6.2	Transformations of Metric Perturbations	15
6.3	Gauge Freedom in the Synchronous Gauge	16
6.4	Bardeen Potentials: Gauge-Invariant Metric Variables	16
6.5	Gauge-Invariant Matter Variables	17
6.6	Physical Interpretation and Applications	17
7	Boltzmann and Fluid Equations for Matter and Radiation	18
7.1	Phase-Space Distribution and the Boltzmann Equation	18
7.2	Fluid Equations for Cold Dark Matter and Baryons	18
7.3	Boltzmann Hierarchy for Photons	19
7.4	Massless and Massive Neutrinos	19
7.5	Tight-Coupling Limit and Initial Conditions	20
8	Evolution of Perturbations Before Horizon Crossing	20
8.1	Superhorizon Solutions in the Synchronous Gauge	20
8.2	Superhorizon Solutions in the Conformal Newtonian Gauge	21
8.3	Gauge Dependence and Matching Conditions	22

8.4	Importance for Numerical Evolution	22
9	Perturbations After Horizon Crossing and Cosmological Observables	22
9.1	Photon–Baryon Oscillations and Acoustic Peaks	23
9.2	Silk Damping and Diffusion Suppression	23
9.3	Neutrino Free-Streaming and its Consequences	24
9.4	Late-Time Integrated Sachs–Wolfe (ISW) Effect	24
9.5	Matter Power Spectrum and Structure Formation	24
9.6	Connecting Theory to Observations	25
10	Numerical Implementation and Einstein–Boltzmann Codes	26
10.1	Motivation for Numerical Integration	26
10.2	The Einstein–Boltzmann System	26
10.3	Gauge Choice and Stability	26
10.4	Initial Conditions	27
10.5	CMBFAST, CAMB, and CLASS	27
11	Conclusion and Future Outlook	28

1. Introduction

Cosmological perturbation theory plays a central role in connecting theoretical predictions from early-universe physics to current astrophysical and cosmological observations. By introducing small perturbations around the homogeneous and isotropic Friedmann–Lemaître–Robertson–Walker (FLRW) background, we are able to model the evolution of cosmic inhomogeneities, from quantum fluctuations generated during inflation to the anisotropies in the cosmic microwave background (CMB) and the large-scale distribution of galaxies.

The development of this theory traces back to the pioneering work of Lifshitz in 1946 [3], who first studied linear perturbations of the FLRW metric in what is now known as the synchronous gauge. Subsequent refinements were necessary to address gauge ambiguities—coordinate-dependent effects that do not correspond to physical observables. The introduction of gauge-invariant variables by Bardeen [2] and the formulation of the conformal Newtonian (or longitudinal) gauge have since become standard tools for analyzing perturbations in both analytic and numerical studies.

In the modern era, cosmological perturbation theory underpins the interpretation of high-precision measurements from CMB experiments (e.g., WMAP, Planck), baryon acoustic oscillations, redshift-space distortions, and weak gravitational lensing. The success of the Λ CDM model, which posits a Universe dominated by cold dark matter and dark energy, relies heavily on linear theory to compute the power spectra of temperature and polarization anisotropies in the CMB, as well as matter clustering statistics.

Gauge choice remains a crucial issue in perturbation theory. While gauge-invariant formulations offer elegance and generality, specific gauge choices often yield computational and interpretive clarity. The synchronous gauge, for instance, is favored in N-body simulations and analytic treatments of cold dark matter (CDM) evolution, despite requiring additional conditions to remove spurious gauge modes. In contrast, the conformal Newtonian gauge aligns closely with Newtonian intuition and observationally relevant quantities, especially in scalar perturbations.

This report presents a detailed treatment of linear perturbation theory in both the synchronous and conformal Newtonian gauges, following the formalism developed by Ma and Bertschinger [1]. We systematically derive the perturbed Einstein, Boltzmann, and fluid equations for a Universe composed of photons, baryons, CDM, and neutrinos—both massless and massive. In doing so, we clarify the evolution of scalar metric perturbations, the impact of gauge transformations, and the formulation of physically meaningful initial conditions.

The report is organized as follows. Section 2 introduces the FLRW background and the fundamental symmetries of cosmological models. Section 3 discusses the classification of perturbations into scalar, vector, and tensor modes, and the general decomposition of

the metric perturbations. Section 4 and Section 5 develop the perturbation theory in the synchronous and conformal Newtonian gauges, respectively. Section 6 addresses gauge transformations and the construction of gauge-invariant quantities. In Sections 7 and 8, we derive the evolution equations and examine solutions before and after horizon crossing. Section 9 discusses applications to CMB anisotropies, matter power spectra, and the effect of neutrinos. Finally, Section 10 outlines the numerical implementation of these equations, and Section 11 concludes with a summary and outlook.

Throughout this work, we adopt natural units ($c = \hbar = 1$) and assume a spatially flat Universe unless otherwise stated. We use conformal time η and denote derivatives with respect to η by primes ($'$). Metric signature is $(-, +, +, +)$.

2. The FLRW Background and Cosmological Symmetries

To study cosmological perturbations within the framework of general relativity, we begin by introducing the background solution — a spatially homogeneous and isotropic universe described by the Friedmann–Lemaître–Robertson–Walker (FLRW) metric. This metric forms the geometric foundation on which small perturbations are later introduced.

2.1. FLRW Metric in Conformal Time

In conformal coordinates, the FLRW metric takes the form:

$$ds^2 = a^2(\eta) \left(-d\eta^2 + \gamma_{ij} dx^i dx^j \right), \quad (2.1)$$

where η is the conformal time defined by $d\eta = dt/a(t)$, $a(\eta)$ is the scale factor, and γ_{ij} is the spatial 3-metric of constant curvature ($k = 0, \pm 1$).

For a spatially flat universe ($k = 0$), which is observationally favored, $\gamma_{ij} = \delta_{ij}$, and the metric simplifies to:

$$ds^2 = a^2(\eta) \left(-d\eta^2 + dx^i dx^i \right). \quad (2.2)$$

This conformal form is particularly useful for analyzing light propagation and perturbations, since it is conformally related to Minkowski spacetime.

2.2. Perfect Fluid Energy-Momentum Tensor

We assume the universe is filled with a perfect fluid. The energy-momentum tensor is:

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + P g^{\mu\nu}, \quad (2.3)$$

where ρ is the energy density, P the pressure, and u^μ the four-velocity satisfying $u^\mu u_\mu = -1$. In the background FLRW solution, the fluid is comoving, with $u^\mu = (\frac{1}{a}, 0, 0, 0)$.

The total energy-momentum tensor is obtained by summing over contributions from all components of the cosmic fluid: photons, baryons, cold dark matter (CDM), massless and massive neutrinos, and possibly a cosmological constant.

2.3. Background Einstein Equations

To describe the large-scale dynamics of the Universe, we begin with Einstein's field equations applied to a homogeneous and isotropic background spacetime described by the FLRW metric. In conformal time η , the metric takes the form:

$$ds^2 = a^2(\eta) \left(-d\eta^2 + dx^i dx^i \right), \quad (2.4)$$

assuming spatial flatness ($k = 0$). The dynamics of the scale factor $a(\eta)$ are governed by the Einstein equations:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (2.5)$$

In the FLRW background, the stress-energy tensor is that of a perfect fluid:

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + P g^{\mu\nu}, \quad (2.6)$$

where ρ is the energy density, P is the pressure, and u^μ is the four-velocity of the fluid, satisfying $u^\mu u_\mu = -1$. In the comoving frame, $u^\mu = (1/a, 0, 0, 0)$.

Substituting this into Einstein's equations, we obtain the two Friedmann equations in conformal time:

$$\mathcal{H}^2 \equiv \left(\frac{a'}{a}\right)^2 = \frac{8\pi G}{3} a^2 \rho, \quad (2.7)$$

$$\mathcal{H}' = \frac{a''}{a} - \mathcal{H}^2 = -\frac{4\pi G}{3} a^2 (\rho + 3P), \quad (2.8)$$

where a prime denotes derivative with respect to conformal time η , and $\mathcal{H} = a'/a$ is the conformal Hubble parameter.

These equations describe how the scale factor evolves with time, depending on the total energy content of the Universe.

Additionally, the conservation of energy-momentum, given by the vanishing covariant divergence of the stress-energy tensor,

$$\nabla_\mu T^{\mu\nu} = 0, \quad (2.9)$$

leads to the continuity equation for the background energy density:

$$\rho' + 3\mathcal{H}(\rho + P) = 0. \quad (2.10)$$

If the Universe is composed of multiple non-interacting fluids, each component obeys its own continuity equation independently.

2.4. Equation of State and Cosmological Epochs

The relationship between pressure and energy density for a cosmic fluid is characterized by its equation of state:

$$P = w\rho, \quad (2.11)$$

where w is the equation of state parameter. Different components of the Universe are characterized by different values of w :

- **Radiation (photons and massless neutrinos):** $w = \frac{1}{3} \Rightarrow P = \frac{1}{3}\rho$
- **Non-relativistic matter (baryons, cold dark matter):** $w \approx 0 \Rightarrow P \approx 0$
- **Cosmological constant (vacuum energy):** $w = -1 \Rightarrow P = -\rho$

The form of the continuity equation (2.10) then leads to different scaling behaviors for $\rho(a)$ depending on w :

$$\rho(a) \propto a^{-3(1+w)}. \quad (2.12)$$

This gives the following evolution laws:

- For radiation: $\rho_r \propto a^{-4}$
- For matter: $\rho_m \propto a^{-3}$
- For vacuum energy: $\rho_\Lambda = \text{constant}$

Cosmic Expansion History

The dominant component of the Universe changes over time, leading to distinct cosmological epochs, each with a characteristic evolution of the scale factor:

- **Radiation-dominated era** ($a \ll a_{\text{eq}}$):

The energy density is dominated by relativistic species ($w = 1/3$). Solving the Friedmann equation leads to:

$$a(\eta) \propto \eta.$$

- **Matter-dominated era** ($a_{\text{eq}} \ll a \ll a_\Lambda$):

Non-relativistic matter dominates ($w \approx 0$), and the scale factor evolves as:

$$a(\eta) \propto \eta^2.$$

- **Dark energy-dominated era** ($a \gtrsim a_\Lambda$):

Vacuum energy (cosmological constant) dominates ($w = -1$). The Universe undergoes accelerated expansion:

$$a(t) \propto e^{Ht}, \quad \text{or} \quad a(\eta) \propto \frac{-1}{\eta}, \quad \eta < 0.$$

These epochs form the backdrop against which cosmological perturbations evolve. The time dependence of the scale factor $a(\eta)$ and the background densities $\rho(\eta)$ directly affect how perturbations grow or decay in different eras, which is essential when solving the Einstein and Boltzmann equations in later sections.

2.5. Relevance to Perturbation Theory

The scale factor $a(\eta)$ and background quantities like $\rho(\eta)$ determine how perturbations evolve. For example, gravitational potentials decay during radiation domination, affecting CMB anisotropies. Spatial flatness ($k = 0$) simplifies perturbation expansions, as eigenfunctions of the Laplacian become plane waves.

The background forms the "zero-order" universe — a dynamical but symmetric space-time — onto which perturbations are added. Understanding its evolution is essential before introducing inhomogeneities in subsequent sections.

3. Classification of Perturbations and Metric Decomposition

In linear cosmological perturbation theory, we analyze small deviations from the homogeneous and isotropic FLRW background. Due to the high degree of symmetry of the FLRW metric (homogeneity and isotropy in space), the linearized Einstein equations permit a natural decomposition of perturbations into three categories: **scalar**, **vector**, and **tensor** modes.

This classification is based on the transformation properties of the perturbations under 3-dimensional spatial coordinate transformations (rotations and translations), and it ensures that, at linear order, these three classes evolve independently. This decoupling is crucial for simplifying the Einstein-Boltzmann system.

3.1. General Metric Perturbation Structure

We denote the total metric as a perturbed expansion around the FLRW background:

$$g_{\mu\nu}(x, \eta) = \bar{g}_{\mu\nu}(\eta) + h_{\mu\nu}(x, \eta), \quad (3.1)$$

where $\bar{g}_{\mu\nu} = a^2(\eta)\eta_{\mu\nu}$ is the conformal FLRW metric (with $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$), and $h_{\mu\nu}$ represents the first-order perturbation.

Since $h_{\mu\nu}$ is a symmetric 4×4 tensor, it has 10 independent components. These components can be split into:

- **4 scalar degrees of freedom:** ϕ , ψ , B , and E ;
- **4 vector degrees of freedom:** from the divergence-free parts of $B_i^{(V)}$ and $E_i^{(V)}$;
- **2 tensor degrees of freedom:** from the transverse-traceless tensor $h_{ij}^{(T)}$.

3.2. Metric Perturbation Decomposition: Scalar, Vector, and Tensor

The most general scalar, vector, and tensor (SVT) decomposition of the perturbed metric in conformal time is:

$$ds^2 = a^2(\eta) \left[- (1 + 2\phi) d\eta^2 + 2(B_{,i} + B_i^{(V)}) d\eta dx^i + \left((1 - 2\psi)\delta_{ij} + 2E_{,ij} + E_{i,j}^{(V)} + E_{j,i}^{(V)} + h_{ij}^{(T)} \right) dx^i dx^j \right]. \quad (3.2)$$

Here:

- ϕ and ψ are scalar potentials representing Newtonian-like gravitational perturbations.
- B and E are scalar functions that appear in the shift and spatial metric perturbations.
- $B_i^{(V)}$ and $E_i^{(V)}$ are divergence-free vector fields (transverse: $\partial^i B_i^{(V)} = 0$), accounting for vorticity.
- $h_{ij}^{(T)}$ is a transverse-traceless tensor satisfying:

$$\partial^i h_{ij}^{(T)} = 0, \quad h_i^{i(T)} = 0,$$

and represents gravitational waves.

Each of these SVT components evolves independently because they transform irreducibly under spatial rotations. Therefore, one can solve their equations separately in linear theory, significantly simplifying the analysis.

3.3. Harmonic Expansion in Flat Space

In a spatially flat background (which we assume throughout, $k = 0$), spatial perturbations can be expanded in Fourier modes. The eigenfunctions of the Laplacian operator are plane waves:

$$\nabla^2 Q(\vec{x}) = -k^2 Q(\vec{x}), \quad Q(\vec{x}) = e^{i\vec{k}\cdot\vec{x}}. \quad (3.3)$$

Using the scalar eigenfunctions $Q(\vec{x})$, we can construct:

$$Q_i^{(S)} = -ik_i Q(\vec{x}), \quad (3.4)$$

$$Q_{ij}^{(S)} = \left(k_i k_j - \frac{1}{3} \delta_{ij} k^2 \right) Q(\vec{x}), \quad (3.5)$$

to represent scalar-vector-tensor basis functions for perturbation expansion.

Similar harmonic functions exist for vector and tensor modes using transverse vector and tensor eigenfunctions. These are essential for the Boltzmann hierarchy and gravitational wave analysis.

3.4. Fourier Decomposition of Perturbations

Each perturbation variable (e.g., $h_{\mu\nu}$, $\delta\rho$, θ , etc.) is Fourier transformed:

$$h_{\mu\nu}(x, \eta) = \int \frac{d^3k}{(2\pi)^3} \sum_{n=S,V,T} h_{\mu\nu}^{(n)}(\vec{k}, \eta) Q^{(n)}(\vec{k}, \vec{x}), \quad (3.6)$$

where n indexes the scalar (S), vector (V), and tensor (T) modes.

Because different $\vec{k}$ -modes evolve independently in linear theory, one can treat each mode separately, reducing the Einstein and Boltzmann equations to a system of ODEs for each wavenumber. This is the foundation for numerical codes like **CAMB** and **CLASS**.

3.5. Interpretation of the SVT Modes

- **Scalars:** Govern structure formation and dominate in standard cosmological models. They couple to energy density perturbations, pressure, and potential wells.
- **Vectors:** Represent vorticity and are typically sourced by decaying modes. In inflationary cosmology, vector perturbations are negligible.
- **Tensors:** Gravitational waves, which leave imprints in the CMB B-mode polarization. They arise from quantum fluctuations during inflation.

This decomposition provides the groundwork for analyzing cosmological perturbations in any gauge. In the following sections, we apply this formalism to specific gauge choices: the synchronous gauge and the conformal Newtonian gauge.

In the next sections, we will study how these perturbations are treated in specific gauges, beginning with the synchronous gauge.

4. The Synchronous Gauge

The synchronous gauge is one of the earliest coordinate choices adopted in cosmological perturbation theory. Introduced by Lifshitz (1946), it simplifies the perturbed metric by eliminating the time-time and time-space components of the metric perturbation. However, it retains some residual gauge freedom, requiring additional conditions to fully fix the coordinate system.

4.1. Gauge Conditions and Metric Form

In the synchronous gauge, the metric perturbations satisfy the conditions:

$$h_{00} = 0, \quad h_{0i} = 0. \quad (4.1)$$

This leads to a simplified form of the perturbed FLRW metric in conformal time:

$$ds^2 = a^2(\eta) \left[-d\eta^2 + (\delta_{ij} + h_{ij}) dx^i dx^j \right], \quad (4.2)$$

where h_{ij} encodes the spatial part of the metric perturbation.

For scalar perturbations, h_{ij} can be decomposed in Fourier space as:

$$h_{ij}(\vec{k}, \eta) = \hat{k}_i \hat{k}_j h(\vec{k}, \eta) + \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) 6\eta(\vec{k}, \eta), \quad (4.3)$$

where h and η are the two scalar metric perturbation variables, and $\hat{k}_i = k_i/|\vec{k}|$.

Note: The symbol η used here refers to a scalar perturbation variable, *not* to be confused with the conformal time variable, which also appears throughout this report.

4.2. Residual Gauge Freedom

Although the synchronous gauge sets $h_{00} = h_{0i} = 0$, it does not completely eliminate gauge degrees of freedom. There remain two unphysical (non-dynamical) modes:

- A constant mode in h corresponding to uniform spatial translations;
- A linearly growing mode in h corresponding to rescaling of spatial coordinates.

To remove these residual gauge modes, a common practice is to anchor the coordinate system to the cold dark matter (CDM) rest frame by setting its peculiar velocity to zero:

$$\theta_c = 0. \quad (4.4)$$

4.3. Perturbed Einstein Equations in Synchronous Gauge

The Einstein equations yield evolution equations for the scalar metric perturbations h and η in Fourier space:

$$k^2 \eta - \frac{1}{2} \mathcal{H} h' = 4\pi G a^2 \delta T^0_0, \quad (4.5)$$

$$k^2 \eta' = 4\pi G a^2 (\bar{\rho} + \bar{P}) \theta, \quad (4.6)$$

$$h'' + 2\mathcal{H} h' - 2k^2 \eta = -8\pi G a^2 \delta T^i_i, \quad (4.7)$$

$$h'' + 6\eta'' + 2\mathcal{H}(h' + 6\eta') - 2k^2 \eta = -24\pi G a^2 (\bar{\rho} + \bar{P}) \sigma, \quad (4.8)$$

where:

- $\mathcal{H} = a'/a$ is the conformal Hubble parameter,
- θ is the velocity divergence,

- σ is the anisotropic stress,
- δT_0^0 and δT_i^i are perturbations to the energy and pressure densities.

4.4. Advantages and Limitations

The synchronous gauge is particularly convenient for analyzing cold dark matter, since CDM is often modeled as a pressureless fluid at rest in this frame. It is also the default choice in many Boltzmann codes (e.g., CAMB).

However, its main limitation is the presence of unphysical gauge modes that must be carefully eliminated. Physical interpretation of perturbation variables can also be less intuitive compared to other gauges.

In the next section, we explore the conformal Newtonian (or longitudinal) gauge, which offers a gauge-fixed and observationally transparent framework.

5. The Conformal Newtonian Gauge

The conformal Newtonian gauge—also referred to as the longitudinal gauge—is a widely used gauge in cosmological perturbation theory due to its clear physical interpretation and its elimination of gauge ambiguities. It is particularly useful for scalar perturbations and for comparison with Newtonian intuition.

5.1. Gauge Conditions and Metric Form

In this gauge, the metric is fully specified by two scalar potentials, ϕ and ψ . The off-diagonal space-time and the scalar spatial shear terms are set to zero:

$$B = 0, \quad E = 0. \tag{5.1}$$

The perturbed FLRW metric becomes:

$$ds^2 = a^2(\eta) \left[-(1 + 2\psi)d\eta^2 + (1 - 2\phi)\delta_{ij}dx^i dx^j \right], \tag{5.2}$$

where:

- ψ is the Newtonian gravitational potential,
- ϕ describes spatial curvature perturbations.

5.2. Physical Interpretation

The conformal Newtonian gauge is fully gauge-fixed, meaning that all residual coordinate freedom is removed. This makes ϕ and ψ true gauge-invariant quantities under linear theory.

- ψ determines gravitational redshift, lensing, and time delay effects.
- ϕ controls the motion of test particles and curvature of space.

In the absence of anisotropic stress (e.g., in single-component perfect fluids), general relativity predicts:

$$\phi = \psi. \quad (5.3)$$

5.3. Perturbed Einstein Equations in This Gauge

The linearized Einstein equations for scalar perturbations in Fourier space become:

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = 4\pi Ga^2\delta\rho, \quad (5.4)$$

$$k^2(\phi' + \mathcal{H}\psi) = 4\pi Ga^2(\bar{\rho} + \bar{P})\theta, \quad (5.5)$$

$$\phi'' + \mathcal{H}(2\phi' + \psi') + (2\mathcal{H}' + \mathcal{H}^2)\psi = 4\pi Ga^2\delta P + \frac{1}{3}k^2(\phi - \psi), \quad (5.6)$$

$$k^2(\phi - \psi) = 12\pi Ga^2(\bar{\rho} + \bar{P})\sigma, \quad (5.7)$$

where:

- $\delta\rho$ and δP are density and pressure perturbations,
- θ is the velocity divergence,
- σ is the anisotropic stress,
- $\mathcal{H} = a'/a$ is the conformal Hubble parameter.

5.4. Advantages of the Gauge

This gauge offers several key advantages:

- No residual gauge freedom—results are directly physical.
- Variables ϕ and ψ have clear physical interpretations.
- Particularly useful for late-time Universe and scalar-mode analysis.
- Often used in relativistic corrections to Newtonian N-body simulations.

Because of these advantages, the conformal Newtonian gauge is commonly used in both analytic calculations and numerical codes that aim to predict observables such as weak lensing, galaxy clustering, and CMB anisotropies.

In the next section, we derive the Boltzmann and fluid equations for different species of particles—photons, baryons, CDM, and neutrinos—coupled to the metric perturbations in both gauges.

6. Gauge Transformations and Gauge-Invariant Variables

A central challenge in cosmological perturbation theory is the presence of *gauge freedom*—the freedom to choose coordinates when splitting the perturbed universe into background and fluctuations. This leads to apparent perturbations that are artifacts of the coordinate system, not physical inhomogeneities. In this section, we introduce gauge transformations, show how perturbation variables change under them, and define gauge-invariant quantities that capture the true physical content.

6.1. First-Order Gauge Transformations

Under an infinitesimal coordinate transformation:

$$x^\mu \rightarrow \tilde{x}^\mu = x^\mu + \xi^\mu, \quad (6.1)$$

where $\xi^\mu = (\alpha, \partial^i \beta)$ defines the gauge generator, the metric and matter perturbations transform accordingly. This transformation induces changes in the perturbation variables, even though the underlying spacetime remains unchanged.

6.2. Transformations of Metric Perturbations

We consider the scalar perturbations to the FLRW metric in conformal time:

$$ds^2 = a^2(\eta) \left[-(1 + 2\phi) d\eta^2 + 2\partial_i B d\eta dx^i + ((1 - 2\psi)\delta_{ij} + 2\partial_i \partial_j E) dx^i dx^j \right]. \quad (6.2)$$

Under the gauge transformation $\xi^\mu = (\alpha, \partial^i \beta)$, the scalar perturbations transform as:

$$\tilde{\phi} = \phi - \mathcal{H}\alpha - \alpha', \quad (6.3)$$

$$\tilde{\psi} = \psi + \mathcal{H}\alpha, \quad (6.4)$$

$$\tilde{B} = B + \alpha - \beta', \quad (6.5)$$

$$\tilde{E} = E - \beta, \quad (6.6)$$

where a prime denotes a derivative with respect to conformal time η , and $\mathcal{H} = a'/a$ is the conformal Hubble parameter.

These transformations show explicitly that quantities like ϕ and ψ are not gauge-invariant in general.

6.3. Gauge Freedom in the Synchronous Gauge

The synchronous gauge is defined by setting:

$$\phi = 0, \quad B = 0. \quad (6.7)$$

However, this does not fully eliminate gauge freedom. In particular, there remains residual freedom in choosing α and β such that the synchronous conditions are preserved. This residual gauge freedom allows unphysical modes (e.g., linearly growing solutions in h and η) that must be removed by fixing additional conditions—typically by demanding that cold dark matter has vanishing peculiar velocity: $\theta_c = 0$.

6.4. Bardeen Potentials: Gauge-Invariant Metric Variables

One approach to overcoming gauge ambiguity is to construct variables that are explicitly gauge-invariant. Two such variables were introduced by Bardeen and are widely used:

$$\Phi \equiv \phi + \mathcal{H}(B - E') + (B - E)', \quad (6.8)$$

$$\Psi \equiv \psi - \mathcal{H}(B - E'). \quad (6.9)$$

These combinations are invariant under the scalar-type gauge transformations. In the conformal Newtonian gauge (where $B = E = 0$), they reduce to:

$$\Phi = \phi, \quad \Psi = \psi.$$

Hence, in this gauge, the potentials ϕ and ψ directly represent physical, gauge-invariant gravitational potentials.

6.5. Gauge-Invariant Matter Variables

Analogous gauge-invariant combinations can be constructed for matter perturbations. For example, the gauge-invariant density contrast in the comoving gauge is:

$$\Delta \equiv \delta + 3(1 + w)\mathcal{H}(v - B), \quad (6.10)$$

where $w = P/\rho$ is the equation-of-state parameter and v is the velocity potential.

Similarly, the curvature perturbation on uniform density hypersurfaces is given by:

$$\zeta \equiv -\psi - \frac{\mathcal{H}}{\rho'}\delta\rho, \quad (6.11)$$

and the comoving curvature perturbation is:

$$\mathcal{R} \equiv \psi - \frac{\mathcal{H}}{\rho + P}\delta q, \quad (6.12)$$

where δq is the momentum potential related to the fluid velocity.

These quantities are conserved on superhorizon scales for adiabatic perturbations, making them useful for setting initial conditions and tracing the evolution of perturbations across different cosmological epochs.

6.6. Physical Interpretation and Applications

Gauge-invariant variables like Φ , Ψ , ζ , and $\mathcal{R}$ are essential tools in cosmology. They:

- Provide consistent physical interpretation independent of coordinate choices.
- Are used directly in theoretical predictions of the Cosmic Microwave Background (CMB) and large-scale structure.
- Serve as boundary conditions for numerical Boltzmann solvers.
- Remain conserved (under certain conditions), offering insights into early universe physics such as inflation.

In summary, while gauge freedom complicates the interpretation of cosmological perturbations, the construction of gauge-invariant variables restores physical clarity. This formalism allows consistent comparison between analytical results, numerical codes, and observational data.

7. Boltzmann and Fluid Equations for Matter and Radiation

The evolution of cosmological perturbations is governed by the interplay between gravity (through metric perturbations) and the behavior of matter and radiation, described by the Boltzmann and fluid equations. Different species—cold dark matter (CDM), baryons, photons, and neutrinos—respond differently due to their distinct physical properties.

7.1. Phase-Space Distribution and the Boltzmann Equation

For relativistic particles, the distribution function $f(x^i, q^i, \eta)$ evolves according to the collisionless Boltzmann equation:

$$\frac{Df}{D\eta} = \frac{\partial f}{\partial \eta} + \frac{dx^i}{d\eta} \frac{\partial f}{\partial x^i} + \frac{dq^i}{d\eta} \frac{\partial f}{\partial q^i} = C[f], \quad (7.1)$$

where q^i is the comoving momentum, and $C[f]$ is the collision term. For collisionless species like massless neutrinos after decoupling, $C[f] = 0$.

We expand f around the homogeneous distribution:

$$f(x^i, q^i, \eta) = \bar{f}(q) [1 + \Psi(x^i, q^i, \eta)], \quad (7.2)$$

where Ψ encodes the perturbation.

7.2. Fluid Equations for Cold Dark Matter and Baryons

For CDM, modeled as a pressureless fluid, the continuity and Euler equations in synchronous gauge are:

$$\delta'_c = -\theta_c - \frac{1}{2}h', \quad (7.3)$$

$$\theta'_c = -\mathcal{H}\theta_c, \quad (7.4)$$

where δ_c is the CDM density contrast, θ_c is its velocity divergence, and h is the synchronous metric perturbation.

For baryons, which interact with photons via Thomson scattering before recomb-

nation:

$$\delta'_b = -\theta_b - \frac{1}{2}h', \quad (7.5)$$

$$\theta'_b = -\mathcal{H}\theta_b + c_s^2 k^2 \delta_b + R^{-1} a n_e \sigma_T (\theta_\gamma - \theta_b), \quad (7.6)$$

where c_s is the baryon sound speed, $R = \frac{3\rho_b}{4\rho_\gamma}$, n_e is the electron density, and σ_T is the Thomson cross-section.

7.3. Boltzmann Hierarchy for Photons

Photons require a Boltzmann hierarchy due to their angular dependence. Define:

$$\Theta(\vec{k}, \mu, \eta) = \sum_{\ell=0}^{\infty} (-i)^\ell (2\ell+1) \Theta_\ell(\vec{k}, \eta) P_\ell(\mu), \quad (7.7)$$

where $\mu = \hat{k} \cdot \hat{n}$ is the cosine of the angle between wavevector and photon direction.

The photon hierarchy in synchronous gauge is:

$$\Theta'_0 = -\frac{1}{3}\theta_\gamma - \frac{1}{6}h', \quad (7.8)$$

$$\Theta'_1 = \frac{k}{3}(\Theta_0 - 2\Theta_2) + a n_e \sigma_T (\theta_b - \theta_\gamma), \quad (7.9)$$

$$\Theta'_2 = \frac{k}{5}(2\Theta_1 - 3\Theta_3) - a n_e \sigma_T \left(\Theta_2 - \frac{1}{10}\Pi \right), \quad (7.10)$$

$$\Theta'_\ell = \frac{k}{2\ell+1}[\ell\Theta_{\ell-1} - (\ell+1)\Theta_{\ell+1}] - a n_e \sigma_T \Theta_\ell \quad (\ell \geq 3), \quad (7.11)$$

where $\Pi = \Theta_2 + \Theta_2^{(P)}$ includes polarization contributions.

7.4. Massless and Massive Neutrinos

Massless neutrinos evolve like photons, but without collisions:

$$\Psi' + ik\mu\Psi = 0, \quad (7.12)$$

leading to a similar hierarchy of multipoles.

Massive neutrinos require integrating over momentum:

$$\epsilon = \sqrt{q^2 + a^2 m^2}, \quad (7.13)$$

$$\frac{dq}{d\eta} = -q \frac{\partial \phi}{\partial \eta} - \epsilon \hat{n}^i \partial_i \psi. \quad (7.14)$$

The Boltzmann equation is then expanded in multipoles and integrated over q to compute $\delta\rho$, δP , θ , and σ .

7.5. Tight-Coupling Limit and Initial Conditions

Before recombination, photons and baryons are tightly coupled. In this limit, photon higher multipoles ($\ell \geq 2$) are negligible, and $\theta_\gamma \approx \theta_b$.

Initial conditions are typically set in the superhorizon regime ($k\eta \ll 1$) with adiabatic or isocurvature modes. For adiabatic initial conditions, the density contrasts of all species are related:

$$\frac{\delta_\gamma}{4} = \frac{\delta_\nu}{4} = \frac{\delta_b}{3} = \frac{\delta_c}{3}. \quad (7.15)$$

In the next section, we examine how these perturbations evolve before horizon crossing in different epochs of the Universe.

8. Evolution of Perturbations Before Horizon Crossing

Perturbation modes with comoving wavelength much larger than the Hubble radius (i.e., modes satisfying $k\eta \ll 1$) are said to lie in the *superhorizon regime*. In this regime, causal microphysics (e.g., pressure forces, collisions) are ineffective, and the evolution of perturbations is governed solely by gravity and the background expansion. This allows for analytical solutions, which are used to initialize full numerical evolution later.

8.1. Superhorizon Solutions in the Synchronous Gauge

In the synchronous gauge, the metric perturbation satisfies the gauge conditions:

$$h_{00} = 0, \quad h_{0i} = 0.$$

However, this gauge leaves residual gauge freedom, corresponding to spurious modes. To eliminate them, one typically adopts the *CDM rest frame condition*, setting the CDM velocity divergence to zero:

$$\theta_c = 0.$$

Under adiabatic initial conditions during the radiation-dominated era, the dominant

growing mode solutions take the form:

$$h(\eta, k) = C(k\eta)^2, \quad (49)$$

$$\eta_{\text{metric}}(\eta, k) = \frac{C}{6}(k\eta)^2, \quad (50)$$

$$\delta_c = -\frac{1}{2}h = -\frac{C}{2}(k\eta)^2, \quad (51)$$

$$\delta_\gamma = \delta_\nu = -\frac{2}{3}h = -\frac{2C}{3}(k\eta)^2, \quad (52)$$

$$\theta_\gamma = \theta_\nu = -\frac{1}{18}C(k\eta)^3, \quad (53)$$

$$\sigma_\nu = \frac{1}{15}C(k\eta)^2. \quad (54)$$

These relations reflect adiabaticity: all components are perturbed in phase, preserving fixed entropy ratios:

$$\frac{\delta_\gamma}{\delta_c} = \frac{\delta_\nu}{\delta_c} = \frac{4}{3}.$$

8.2. Superhorizon Solutions in the Conformal Newtonian Gauge

In the conformal Newtonian gauge, the metric is given by:

$$ds^2 = a^2(\eta) \left[-(1 + 2\psi)d\eta^2 + (1 - 2\phi)\delta_{ij}dx^i dx^j \right].$$

For adiabatic modes on superhorizon scales, the potentials remain approximately constant:

$$\phi \approx \psi \approx \Phi_0 = \text{const.} \quad (55)$$

The corresponding solutions for density perturbations are:

$$\delta_\gamma = \delta_\nu = -2\Phi_0, \quad (56)$$

$$\delta_c = \delta_b = -\frac{3}{2}\Phi_0, \quad (57)$$

and the velocity divergences evolve as:

$$\theta_\gamma = \theta_\nu = \theta_b = \theta_c = -\frac{k^2\eta}{2}\Phi_0. \quad (58)$$

This gauge yields results with a clear Newtonian interpretation: the gravitational potentials ϕ and ψ act as the scalar potential wells that remain constant until horizon crossing. In the absence of anisotropic stress, general relativity predicts $\phi = \psi$.

8.3. Gauge Dependence and Matching Conditions

Although the variables differ in the two gauges, the physical content of the perturbations is the same. One can relate solutions across gauges using gauge transformation rules.

A gauge-invariant quantity that encapsulates the curvature perturbation is:

$$\mathcal{R} = -\psi - \frac{\mathcal{H}}{\rho + P}\delta q, \quad (59)$$

where $\mathcal{H} = a'/a$ is the conformal Hubble parameter and δq is the momentum potential.

For adiabatic perturbations, $\mathcal{R}$ remains conserved on superhorizon scales. This constancy makes $\mathcal{R}$ a crucial variable for setting initial conditions and for tracing the evolution across different epochs and gauges.

8.4. Importance for Numerical Evolution

These analytic solutions provide the foundation for initializing cosmological Boltzmann solvers. Typically, modes are initialized deep in the superhorizon regime when $k\eta \lesssim 10^{-3}$.

The coherence of the phases across different k -modes (a result of adiabaticity) leads to the formation of acoustic oscillations in the photon-baryon fluid. These oscillations imprint distinctive peaks and troughs in the angular power spectrum of the cosmic microwave background (CMB), observable as:

- The Sachs–Wolfe plateau at large angular scales (low ℓ),
- Acoustic peaks corresponding to horizon-sized modes at recombination,
- Damping tail from photon diffusion (Silk damping) at small scales.

In the next section, we will extend the evolution of these perturbations across horizon crossing and explore how their imprint is transferred to cosmological observables.

9. Perturbations After Horizon Crossing and Cosmological Observables

Once perturbation modes enter the Hubble radius ($k\eta \gtrsim 1$), causal microphysical processes such as pressure support, scattering, and diffusion begin to affect their

evolution. The behavior of these modes depends strongly on the type of cosmic fluid (e.g., radiation, baryons, dark matter, neutrinos), their interactions, and the cosmic epoch.

9.1. Photon–Baryon Oscillations and Acoustic Peaks

Before recombination, photons and baryons are tightly coupled through Thomson scattering. This forms a tightly bound photon–baryon fluid that undergoes acoustic oscillations due to a competition between gravitational infall (compressions) and radiation pressure (rarefactions).

The governing equation for the photon perturbation δ_γ resembles that of a driven, damped harmonic oscillator:

$$\ddot{\delta}_\gamma + \mathcal{H}\dot{\delta}_\gamma + c_s^2 k^2 \delta_\gamma = S(\eta), \quad (9.1)$$

where c_s is the sound speed in the fluid, and $S(\eta)$ is a time-dependent source term involving the gravitational potentials ϕ and ψ .

These acoustic oscillations imprint a series of peaks in the angular power spectrum of the CMB temperature anisotropies, whose features encode information about:

- The baryon-to-photon ratio (affects peak heights—odd peaks are enhanced by compression),
- The total matter content (determines the sound horizon scale and peak locations),
- The primordial curvature perturbations from inflation (sets overall amplitude and spectral tilt).

9.2. Silk Damping and Diffusion Suppression

On small scales (high k), perturbations are damped due to photon diffusion—also known as *Silk damping*. This arises because photons have a finite mean free path, leading to diffusion that mixes hot and cold regions, smoothing out small-scale fluctuations.

The damping introduces an exponential suppression of power for multipoles $\ell \gtrsim 1000$ in the CMB:

$$\Delta_\ell^2 \sim \exp\left(-\frac{k^2}{k_D^2}\right),$$

where k_D is the diffusion (Silk) damping scale. This generates the characteristic *damping tail* in the CMB power spectrum.

9.3. Neutrino Free-Streaming and its Consequences

Massless neutrinos decouple early and free-stream across the Universe. This introduces anisotropic stress, which modifies the evolution of metric perturbations. In particular, it creates a non-zero difference between ϕ and ψ , thereby altering the gravitational driving term in acoustic oscillations:

$$\phi - \psi \neq 0 \quad (\text{due to anisotropic stress}).$$

This slightly suppresses the amplitude and shifts the phase of the acoustic peaks.

Massive neutrinos, which transition from relativistic to non-relativistic behavior, further suppress the growth of matter perturbations on small scales. Their large velocity dispersion leads to free-streaming below a characteristic scale k_{fs} .

The suppression in the matter power spectrum can be approximated as:

$$\frac{\Delta P(k)}{P(k)} \sim -8 \frac{\Omega_\nu}{\Omega_m}, \quad \text{for } k > k_{\text{fs}}. \quad (9.2)$$

This effect has been used to place upper bounds on the sum of neutrino masses using galaxy surveys and CMB data.

9.4. Late-Time Integrated Sachs–Wolfe (ISW) Effect

In a Universe with a significant dark energy component (or Λ), the gravitational potentials ϕ and ψ decay at late times due to the accelerated expansion.

As CMB photons traverse time-varying potentials, they gain or lose energy, resulting in additional temperature anisotropies known as the *Integrated Sachs–Wolfe (ISW) effect*:

$$\left(\frac{\Delta T}{T} \right)_{\text{ISW}} = \int_{\eta_{\text{dec}}}^{\eta_0} (\dot{\phi} + \dot{\psi}) d\eta. \quad (9.3)$$

This effect contributes power to low multipoles ($\ell \lesssim 10$) and can be cross-correlated with large-scale structure surveys to detect signatures of dark energy.

9.5. Matter Power Spectrum and Structure Formation

Cold dark matter (CDM) perturbations begin to grow significantly after matter–radiation equality ($z \sim 3400$), driven by gravitational instability.

The matter power spectrum $P(k)$, defined by:

$$\langle \delta(\vec{k}) \delta^*(\vec{k}') \rangle = (2\pi)^3 \delta_D(\vec{k} - \vec{k}') P(k),$$

describes the variance of matter density fluctuations in Fourier space.

Under adiabatic initial conditions from inflation, the primordial spectrum is nearly scale-invariant:

$$P_{\text{prim}}(k) \propto k^{n_s}, \quad \text{with } n_s \approx 0.96.$$

However, the shape of the observed power spectrum is modified by several effects:

- The transfer function accounting for horizon entry (suppressing small-scale modes),
- Baryonic pressure support and photon drag,
- Neutrino free-streaming (suppressing power at high k),
- Dark energy domination (suppressing late-time growth).

Modern surveys (e.g., BOSS, DESI, Euclid) measure $P(k)$ through galaxy clustering and weak lensing, constraining parameters like Ω_m , H_0 , and the total neutrino mass.

9.6. Connecting Theory to Observations

The evolution of perturbations across cosmic history is imprinted onto observable quantities such as:

- The CMB angular power spectrum C_ℓ (measured by Planck, WMAP, ACT),
- The matter power spectrum $P(k)$ from galaxy redshift surveys (e.g., SDSS, DESI),
- Weak gravitational lensing maps,
- Lyman-alpha forest data and BAO features.

These observables enable precision cosmology and constrain:

- The Hubble constant H_0 and expansion history,
- The matter and baryon densities (Ω_m, Ω_b),
- Neutrino properties (mass, number of species),
- Dark energy equation of state $w(z)$,
- Inflationary parameters (n_s, r, A_s).

In the next section, we describe the numerical implementation of the coupled Einstein–Boltzmann system in codes such as **CAMB** and **CLASS**.

10. Numerical Implementation and Einstein–Boltzmann Codes

The linear evolution of cosmological perturbations involves a coupled system of Einstein equations for the metric and Boltzmann equations for each particle species. Solving these equations numerically is essential for computing the cosmic microwave background (CMB) anisotropies and the matter power spectrum across a wide range of scales and cosmological parameters.

10.1. Motivation for Numerical Integration

While analytical approximations provide valuable intuition for the behavior of perturbations on superhorizon or subhorizon scales, the intermediate regime (e.g., horizon crossing, recombination) and the impact of subtle physical effects (e.g., neutrino anisotropic stress, polarization, reionization) demand full numerical solutions.

Modern observations (e.g., *Planck*, ACT, SPT) require theoretical predictions at percent-level accuracy or better. This necessitates solving the full hierarchy of multipole moments in the Boltzmann equations up to high ℓ .

10.2. The Einstein–Boltzmann System

The set of equations includes:

- Einstein field equations for the metric perturbations (ϕ , ψ or h , η),
- Fluid equations for baryons and CDM,
- Boltzmann equations for photons and neutrinos (massless and massive),
- Coupling terms via gravitational potentials, Thomson scattering, and neutrino anisotropic stress.

Each Boltzmann hierarchy must be truncated at a sufficiently high multipole $\ell_{\max}$ with appropriate closure relations to ensure numerical accuracy.

10.3. Gauge Choice and Stability

Most codes implement the synchronous gauge (as in Ma & Bertschinger) or the conformal Newtonian gauge. Gauge choice affects the numerical behavior and the interpretation of intermediate variables but not the final observable spectra.

Synchronous gauge requires special treatment of gauge modes and careful specification of initial conditions (e.g., fixing CDM velocity to zero). Conformal Newtonian

gauge avoids such issues but requires precise integration of metric evolution due to anisotropic stress.

10.4. Initial Conditions

Initial conditions are typically set deep in the radiation-dominated era, when all relevant perturbation modes are well outside the horizon ($k\eta \ll 1$). For adiabatic initial conditions, all species share coherent perturbations with fixed entropy ratios. Isocurvature modes (e.g., CDM isocurvature, neutrino density isocurvature) can also be studied but are strongly constrained by CMB data.

10.5. CMBFAST, CAMB, and CLASS

Several Einstein–Boltzmann solvers have been developed over the years:

- **CMBFAST**: Developed by Seljak & Zaldarriaga, it pioneered the line-of-sight integration method for efficient computation of the CMB anisotropies.
- **CAMB**: Based on CMBFAST, it is widely used in cosmological parameter estimation pipelines and supports a variety of models, including massive neutrinos, tensor modes, and reionization.
- **CLASS**: A modern code written from scratch, with modular design and extensive support for modified gravity, dark energy, and non-standard models.

These codes numerically evolve the Einstein–Boltzmann system and compute observable quantities such as:

- Temperature (TT), polarization (EE), and cross (TE) CMB angular power spectra,
- Lensing potential and deflection angle power spectra,
- Linear matter power spectrum $P(k)$.

11. Conclusion and Future Outlook

Cosmological perturbation theory provides the essential bridge between theoretical models of the early universe and the wealth of observational data we collect today. By systematically analyzing small deviations from the homogeneous and isotropic FLRW background, we gain insight into the growth of structure, the behavior of dark matter and neutrinos, and the imprint of inflationary physics on the cosmic microwave background (CMB).

In this report, we presented a detailed account of linear perturbation theory in both the synchronous and conformal Newtonian gauges, following the rigorous formalism developed by Ma & Bertschinger (1995). We explored the classification of perturbations into scalar, vector, and tensor modes, the decomposition of metric perturbations, and the derivation of evolution equations for each matter component in the Universe, including cold dark matter, baryons, photons, and both massless and massive neutrinos.

We demonstrated how gauge choice impacts both the theoretical formulation and numerical implementation of the perturbation equations. While the synchronous gauge is widely used in numerical solvers and simulations due to its convenience for cold dark matter, the conformal Newtonian gauge offers an observationally intuitive framework with clear physical interpretations for metric potentials ϕ and ψ .

We discussed the behavior of perturbations before and after horizon crossing, highlighting the analytic superhorizon solutions and the transition to acoustic oscillations, Silk damping, and neutrino free-streaming. These physical processes leave characteristic imprints on the CMB power spectrum and the matter distribution in the late universe.

The numerical integration of the Einstein–Boltzmann system, implemented in codes like CAMB and CLASS, allows us to predict observables with sub-percent accuracy. These tools, combined with data from CMB experiments, galaxy surveys, and weak lensing observations, form the foundation of precision cosmology. They enable stringent constraints on cosmological parameters such as the Hubble constant H_0 , the matter density Ω_m , the neutrino mass sum $\sum m_\nu$, and the nature of dark energy.

Future Outlook

Looking ahead, cosmological perturbation theory will continue to play a central role in upcoming observational missions and theoretical developments. Some promising directions include:

- **CMB Polarization and B-modes:** Next-generation experiments (e.g., CMB-S4, LiteBIRD) aim to detect primordial B-mode polarization patterns, potentially revealing the energy scale of inflation.
- **Large-Scale Structure (LSS) Surveys:** Future surveys like Euclid, Rubin LSST, and DESI will map the distribution of galaxies with unprecedented precision, probing the matter power spectrum and the growth rate of structure.
- **Neutrino Mass Hierarchy:** Cosmology provides complementary constraints on the sum of neutrino masses and may help distinguish between normal and inverted hierarchies.
- **Dark Energy and Modified Gravity:** Precise measurements of the expansion history and structure growth will test models beyond Λ CDM, including evolving dark energy and alternative theories of gravity.
- **Primordial Non-Gaussianity and Isocurvature Modes:** Deviations from Gaussian initial conditions or the presence of isocurvature perturbations would offer deep insights into the physics of the early universe.

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